

AutoPrepAI

Autonomous Multi-Agent Data Science Pipeline
using A2A Communication and MCP Tooling

PROJECT REPORT

An Enterprise-Grade Distributed Autonomous AI Framework using
LLM Orchestration, A2A Communication, FastMCP, JSON-RPC, AsyncIO,
Distributed AI Agents and Autonomous Machine Learning Workflows

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Python, Ollama, FastMCP, HTTPX, AsyncIO, JSON-RPC,
Starlette, Scikit-learn, Pandas, NumPy, SciPy

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Abstract

Modern machine learning workflows require extensive preprocessing, exploratory data analysis, feature engineering, statistical validation, model optimization, and explainability generation before datasets become suitable for predictive intelligence systems. Traditional preprocessing pipelines are rigid, manually configured, tightly coupled, and dependent on hardcoded sequential execution logic that cannot dynamically adapt to heterogeneous datasets.

This project presents **AutoPrepAI**, a distributed autonomous AI-driven data science pipeline integrating Large Language Model (LLM) orchestration, Agent-to-Agent (A2A) communication, and Model Context Protocol (MCP) deterministic tooling for intelligent preprocessing automation. The proposed architecture combines distributed asynchronous AI agents communicating through HTTP and JSON-RPC 2.0 protocols while FastMCP provides deterministic statistical and machine learning execution services.

The system consists of autonomous preprocessing agents including dataset understanding, data quality analysis, missing value handling, encoding, outlier processing, feature engineering, exploratory data analysis, AutoML training, reflection analysis, and explainability report generation. Dynamic runtime discovery mechanisms enable decentralized orchestration without static coupling while the orchestration layer uses LLM reasoning for autonomous preprocessing strategy generation and intelligent workflow planning.

The architecture demonstrates enterprise-grade distributed AI systems engineering principles including asynchronous distributed execution, protocol-driven orchestration, autonomous reasoning, reflection-based optimization, modular service isolation, runtime discovery, memory-aware execution tracking, and explainable analytics generation.

Experimental evaluation demonstrates successful implementation of real A2A communication, FastMCP deterministic tooling, autonomous preprocessing orchestration, distributed service collaboration, runtime discovery, reflection-based optimization, and intelligent machine learning workflow automation.

Keywords : Distributed AI Systems, Autonomous Machine Learning, Agent-to-Agent Communication, Model Context Protocol, FastMCP, Distributed Intelligence, LLM Orchestration, AutoML, Reflection Systems, Async Distributed Computing.

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1 Introduction

Machine learning systems are fundamentally dependent on high-quality preprocessing pipelines before successful model training can occur. Real-world datasets frequently contain missing values, noisy features, skewed distributions, inconsistent schemas, categorical anomalies, outliers, and redundant attributes that require extensive preprocessing before analytical modeling.

Traditional preprocessing systems suffer from multiple limitations:

- Hardcoded execution pipelines
- Monolithic tightly-coupled architectures
- Lack of distributed orchestration
- Poor scalability
- No autonomous reasoning capability
- High manual dependency
- Limited explainability

Recent advancements in Large Language Models (LLMs), distributed AI agents, asynchronous communication frameworks, and protocol-driven orchestration systems have enabled intelligent autonomous infrastructures capable of collaborative execution and adaptive reasoning.

AutoPrepAI is designed as a distributed autonomous AI framework integrating:

- LLM-driven orchestration
- Distributed multi-agent execution
- Agent-to-Agent communication
- FastMCP deterministic tooling
- Reflection-based optimization
- Dynamic runtime discovery
- Autonomous preprocessing orchestration

The architecture demonstrates practical implementation of enterprise-grade distributed AI systems engineering for intelligent machine learning workflow automation.

2 Problem Statement

Modern preprocessing workflows are highly dependent on static rule-based execution systems that cannot dynamically adapt to heterogeneous datasets and evolving analytical requirements.

Existing systems fail to support:

1. Autonomous preprocessing strategy generation
2. Distributed intelligent orchestration
3. Protocol-driven multi-agent communication
4. Runtime-discoverable AI services
5. Reflection-based optimization
6. Explainable preprocessing systems

Traditional preprocessing systems also suffer from:

- High manual intervention requirements
- Poor scalability
- Rigid execution workflows
- Limited fault isolation
- No collaborative AI execution

Therefore, there is a need for a distributed autonomous AI framework capable of intelligent orchestration, scalable preprocessing automation, distributed asynchronous execution, and adaptive machine learning workflow management.

3 Objectives

- Design a distributed autonomous AI architecture.
- Implement Agent-to-Agent communication.
- Integrate FastMCP deterministic tooling.
- Enable LLM-driven orchestration.
- Automate exploratory data analysis workflows.
- Implement reflection-based optimization.
- Generate ML-ready datasets autonomously.
- Produce explainable reports and visualizations.
- Demonstrate enterprise distributed AI engineering principles.

4 Distributed Autonomous Agent Services

AutoPrepAI integrates multiple distributed AI agents operating collaboratively inside an asynchronous execution environment.

Agent	Port	Technical Functionality
DataUnderstandingAgent	8201	Dataset profiling, schema analysis, datatype extraction and metadata generation.
DataQualityAgent	8202	Quality validation, anomaly detection and suspicious feature analysis.
MissingValueAgent	8203	LLM-driven missing value handling and intelligent imputation orchestration.
EncodingAgent	8204	Categorical feature transformation and encoding strategy generation.
OutlierHandlingAgent	8205	Statistical anomaly detection and outlier treatment recommendation workflows.
FeatureEngineeringAgent	8206	Feature generation, transformation planning and feature selection orchestration.
EDAAgent	8207	Exploratory data analysis, visualization generation and statistical interpretation.
AutoMLAgent	8208	Automated model training, evaluation and model optimization workflows.
ReportGenerationAgent	8209	Explainability report generation and artifact aggregation.
MCP Server	8100	FastMCP deterministic statistical execution and machine learning tooling layer.
Orchestrator	8000	Centralized intelligent orchestration and distributed workflow coordination.

Table 1: Distributed Autonomous AI Services

5 System Architecture

AutoPrepAI follows a distributed autonomous multi-agent architecture designed for scalability, modularity, asynchronous execution, protocol-driven orchestration, and intelligent workflow automation.

5.1 Architecture Characteristics

- Distributed Multi-Agent Execution
- LLM-Driven Intelligent Orchestration
- HTTP + JSON-RPC Communication
- FastMCP Deterministic Tooling
- Dynamic Runtime Discovery
- Reflection-Based Optimization
- Asynchronous Distributed Execution
- Modular Service Isolation
- Explainable AI Workflow Generation

5.2 System Architecture Diagram

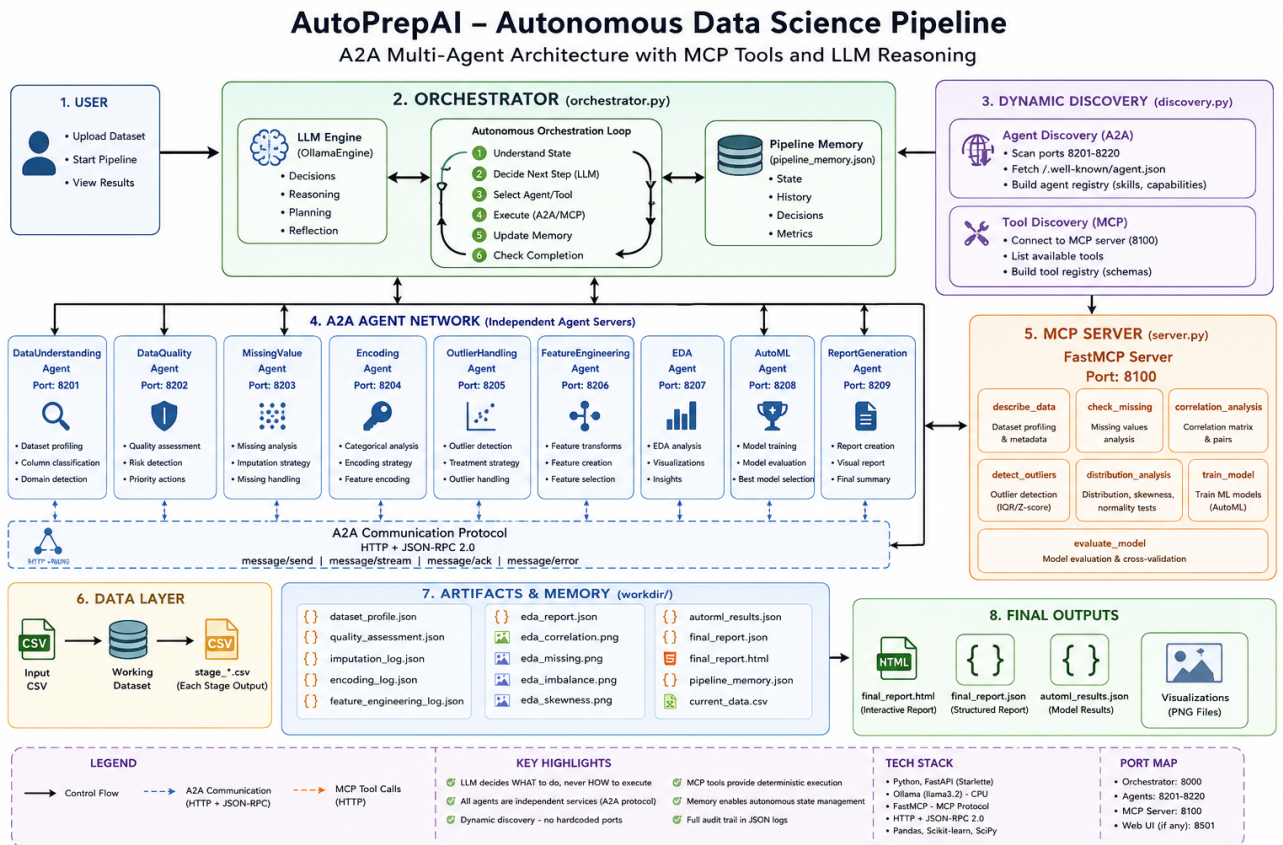


Figure 1: Distributed Autonomous AI Architecture of AutoPrepAI showing LLM-driven orchestration, distributed A2A communication, FastMCP deterministic tooling, autonomous preprocessing, reflection-based optimization, and asynchronous machine learning workflow automation.

The architecture diagram illustrates the complete distributed execution workflow of the AutoPrepAI platform. The orchestration layer dynamically coordinates distributed AI agents through HTTP and JSON-RPC based Agent-to-Agent communication protocols. Autonomous preprocessing agents collab-

oratively perform dataset understanding, quality analysis, missing value handling, encoding, feature engineering, EDA generation, AutoML training, and explainability reporting.

The FastMCP execution layer provides deterministic statistical and machine learning tooling including outlier detection, correlation analysis, distribution analysis, model training, and evaluation services. Dynamic runtime discovery enables decentralized orchestration without static service coupling while the reflection subsystem continuously monitors execution quality and supports autonomous optimization.

6 Agent-to-Agent (A2A) Communication

The project implements real distributed Agent-to-Agent communication using protocol-driven asynchronous messaging architecture.

6.1 A2A Communication Stack

Technology	Purpose
HTTP	Distributed Transport Layer
JSON-RPC 2.0	Protocol Messaging Layer
Starlette	Distributed Agent Server Framework
Uvicorn	Runtime Execution Engine
HTTPX	Asynchronous Communication Client
AsyncIO	Non-Blocking Distributed Execution
Agent Cards	Dynamic Runtime Discovery

Table 2: A2A Communication Stack

6.2 A2A Workflow

1. Orchestrator dynamically discovers active agents.
2. LLM reasoning engine selects next execution target.
3. JSON-RPC protocol requests are generated.
4. HTTP asynchronous communication is initiated.
5. Target distributed agents process requests independently.
6. Structured JSON responses are returned.
7. Pipeline memory and orchestration state are updated.

7 Model Context Protocol (MCP)

AutoPrepAI integrates FastMCP for deterministic execution of statistical and machine learning operations.

7.1 FastMCP Deterministic Tools

- `describe_data`
- `check_missing`
- `detect_outliers`
- `correlation_analysis`
- `distribution_analysis`
- `train_model`
- `evaluate_model`

7.2 MCP Execution Workflow

1. Distributed AI agent invokes MCP tool.
2. FastMCP client generates protocol request.
3. MCP server executes deterministic statistical operation.
4. Structured analytical results are returned.
5. Agent integrates results into orchestration workflow.

8 Distributed Execution Workflow

1. User uploads dataset into orchestration layer.
2. Dynamic runtime discovery identifies active distributed services.
3. LLM-powered orchestrator evaluates dataset state.
4. Intelligent preprocessing strategy is generated autonomously.
5. Distributed AI agents communicate using A2A protocols.
6. FastMCP deterministic statistical tools are invoked.
7. Reflection engine evaluates execution quality.
8. Pipeline memory stores artifacts and execution traces.
9. AutoML training and evaluation workflows execute.
10. Final ML-ready datasets and reports are generated.

9 Results and Performance Analysis

Experimental execution demonstrates successful implementation of distributed autonomous AI orchestration and intelligent preprocessing automation.

9.1 Observed Technical Results

The system successfully performed:

- Autonomous dataset profiling and metadata extraction.
- Distributed preprocessing orchestration using independent AI agents.
- Real A2A communication using HTTP and JSON-RPC protocols.
- Dynamic runtime discovery of agents and services.
- Intelligent missing value handling and preprocessing strategy generation.
- Statistical outlier detection and anomaly analysis.
- Automated exploratory data analysis and visualization workflows.
- Feature engineering and encoding orchestration.
- Autonomous AutoML training and evaluation.
- Reflection-based optimization and execution monitoring.
- Structured explainability report generation.

9.2 Performance Evaluation

Performance Parameter	Observed Status
Distributed Agent Communication	Successful
A2A Protocol Messaging	Operational
FastMCP Tool Invocation	Functional
Dynamic Runtime Discovery	Successful
Autonomous Preprocessing Workflow	Stable
EDA Visualization Generation	Operational
AutoML Pipeline Execution	Successful
Reflection-Based Optimization	Functional
Asynchronous Distributed Execution	Achieved
Structured Report Generation	Successful
Scalable Modular Architecture	Operational

Table 3: Experimental Performance Analysis

9.3 Technical Observations

The distributed architecture significantly improved modularity, scalability, and service isolation. Independent AI agents enabled asynchronous distributed execution while protocol-driven communication reduced tight coupling between services. Reflection-based optimization improved execution reliability and preprocessing adaptability.

The FastMCP deterministic tooling layer successfully separated statistical execution from LLM reasoning, enabling more explainable and controllable preprocessing workflows.

10 Advantages

- Enterprise-grade distributed AI architecture
- Real protocol-driven A2A communication
- Dynamic runtime service discovery
- LLM-driven intelligent orchestration
- FastMCP deterministic statistical tooling
- Reflection-based autonomous optimization
- Explainable preprocessing workflows
- Scalable modular distributed services
- Asynchronous non-blocking execution
- Autonomous machine learning workflow generation

11 Limitations

Despite successful implementation, the system still exhibits several limitations:

- CPU-based LLM inference introduces execution latency.
- Sequential orchestration creates partial bottlenecks.
- Large datasets increase preprocessing overhead.
- Reflection loops increase execution complexity.
- Current deployment is localhost-based.

- Parallel multi-agent planning is limited.
- GPU acceleration is not fully utilized.
- Vector memory retrieval systems are absent.
- Dynamic agent scaling is not fully automated.

12 Future Enhancements

Several future enhancements can further improve the architecture:

- Kubernetes-based cloud-native deployment
- Parallel multi-agent orchestration
- Vector memory and retrieval integration
- Multi-LLM orchestration systems
- Reinforcement learning based planning
- Distributed GPU inference acceleration
- Event-driven streaming pipelines
- Federated distributed AI agents
- Autonomous self-healing workflows
- Advanced explainability analytics
- Real-time adaptive orchestration
- Distributed caching and load balancing

13 Conclusion

AutoPrepAI successfully demonstrates the practical implementation of distributed autonomous AI systems using enterprise-grade orchestration, distributed AI agents, protocol-driven communication, and FastMCP deterministic execution.

The platform integrates:

- Distributed AI agents
- LLM-driven orchestration
- Real A2A communication
- FastMCP deterministic tooling
- Reflection-based optimization
- Autonomous preprocessing workflows
- Explainable machine learning automation

The architecture demonstrates scalable intelligent machine learning workflow automation while maintaining modularity, explainability, asynchronous execution, distributed orchestration, and enterprise distributed systems engineering principles.

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